

UR5e Control Analysis

Part 1B-1 — Joint-Space PID Controller Design and Nominal Benchmark

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September 2026

Abstract

This report documents the design, tuning, and nominal nonlinear benchmark of a joint-space PID controller for the validated UR5e rigid-body model established in Part 1A. I treated tuning as an engineering design study rather than a search for a single numerical optimum. A frozen diagonal inertia approximation provided an interpretable parameterization in terms of nominal bandwidth ω_n , damping ζ , and integral scaling α_i . I then increased bandwidth within a predefined design range, examined damping and integral sensitivity, and selected a final nominal design based on trajectory tracking, terminal error, control effort, and tuning economy.

The selected design, $\omega_n = 20$ rad/s, $\zeta = 3$, and $\alpha_i = 18$ s⁻¹, achieved an overall joint RMSE of 0.4698°, TCP position RMSE of 5.836 mm, and TCP orientation RMSE of 1.528° on the common 4 s quintic benchmark sampled at 2 ms. Maximum torque utilization was only 27.17% on Joint 2. Joint 4 remained the dominant trajectory-tracking limitation, with RMSE 1.098° and maximum error 3.937° despite a final error of only 0.00685°.

An additional inverse-dynamics decomposition was therefore performed to identify the physical origin of the Joint 4 transient. The analysis showed that the Joint 4 torque demand was overwhelmingly gravity-dominated rather than coupling-dominated. At the maximum-error instant, the gravitational contribution was -1.763 N m, while inertial-coupling and Coriolis/centrifugal contributions were only millinewton-metre scale. This identifies Joint 4 as a particularly clear example of the reactive nature of feedback-only PID control when explicit gravity compensation is absent.

The resulting controller provides a deliberately transparent fixed-gain baseline for subsequent computed-torque, LQR, and MPC comparisons. These gains are simulation-tuned gains for the nominal torque-driven rigid-body model, not validated gains for a physical UR5e or its proprietary internal servo.

1 Scope and Benchmark

Part 1A established the nonlinear plant, validated its kinematics and rigid-body dynamics, characterized configuration dependence, and defined the common controller benchmark. This stage asks how far a conventional fixed-gain joint PID can track that benchmark without explicit nonlinear model compensation, which gain changes materially improve performance, and what residual behavior should be carried forward for comparison with model-based controllers.

All controller comparisons use the same nominal conditions:

- 4.0 s quintic joint-space trajectory and 2 ms sample period (500 Hz);
- $q_0 = [0, -90, 90, -90, -90, 0]^\circ$;
- $q_f = [20, -60, 60, -70, -70, 20]^\circ$;
- $|\tau_{1:3}| \leq 150$ N m and $|\tau_{4:6}| \leq 28$ N m;
- nonlinear rigid-body plant integrated by RK4 with zero-order-held control torque.

2 Controller Formulation

The plant is

$$M(q)\ddot{q} + c(q, \dot{q}) + g(q) = \tau. \quad (1)$$

The implemented controller is

$$\tau = K_p e + K_d \dot{e} + K_i \int e dt, \quad e = q_d - q, \quad \dot{e} = \dot{q}_d - \dot{q}. \quad (2)$$

Torque saturation is applied at the adopted simulation limits. Conditional integration provides anti-windup: integration is suppressed when saturation and error direction would drive the command farther beyond the admissible torque.

2.1 Physically interpretable gain parameterization

For an ideal single-joint approximation,

$$M\ddot{e} + K_d \dot{e} + K_p e = 0. \quad (3)$$

Comparison with

$$\ddot{e} + 2\zeta\omega_n \dot{e} + \omega_n^2 e = 0 \quad (4)$$

gives

$$K_p = M\omega_n^2, \quad K_d = 2\zeta M\omega_n. \quad (5)$$

Because the actual UR5e inertia is configuration dependent and coupled, I froze only the diagonal inertia at the benchmark initial configuration,

$$M_d = \text{diag}(M(q_0)), \quad (6)$$

and parameterized the gains as

$$K_p = M_d \omega_n^2, \quad (7)$$

$$K_d = 2\zeta M_d \omega_n, \quad (8)$$

$$K_i = \alpha_i K_p. \quad (9)$$

Thus ω_n and ζ are gain-design parameters, not exact global modal properties of the nonlinear six-axis closed loop.

2.2 Integral-action structural reference

With $z = \int e dt$, the idealized fixed-inertia single-joint PID has

$$M\ddot{e} + K_d \dot{e} + K_p e + K_i z = 0, \quad \dot{z} = e, \quad (10)$$

and characteristic polynomial

$$Ms^3 + K_d s^2 + K_p s + K_i = 0. \quad (11)$$

For positive coefficients, the Routh–Hurwitz condition includes

$$K_i < \frac{K_d K_p}{M}, \quad (12)$$

which becomes

$$\alpha_i < 2\zeta\omega_n. \quad (13)$$

I used this only as a structural reference. It is not a stability certificate for the nonlinear coupled robot with saturation and numerical integration.

3 Tuning Study

3.1 Bandwidth

Initial low-bandwidth PD/PID trials showed substantial error on gravity-loaded and dynamically coupled joints. Increasing ω_n consistently reduced tracking error without approaching the adopted torque limits. At high damping, I swept $\omega_n = 14, 16, 18, 20$ rad/s while holding the integral gain numerically fixed to isolate the P/D bandwidth effect. Overall joint RMSE fell from 2.0726° to 1.5015° , while TCP position RMSE fell from 25.855 to 18.768 mm.

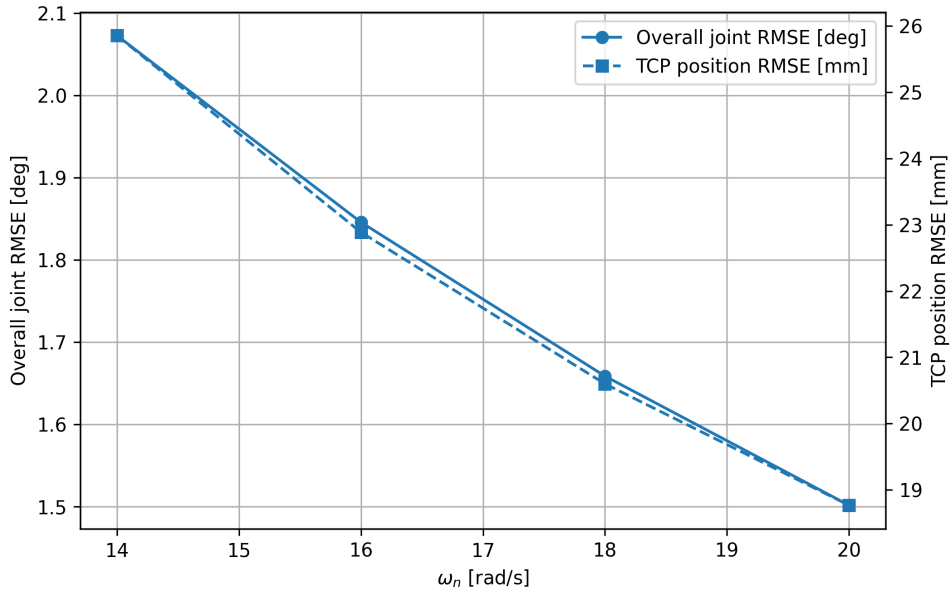


Figure 1: High-damping bandwidth sweep. Integral gain was held numerically fixed during this sweep so the P/D bandwidth effect could be interpreted separately.

I selected 20 rad/s as the upper nominal design bound for this study rather than indefinitely increasing bandwidth to locate a numerical failure boundary. It is therefore a scope boundary, not a mathematical optimum or physical UR5e limit.

3.2 Damping

After the bandwidth and integral action were established, I compared $\zeta = 2$ and $\zeta = 3$ at $\omega_n = 20$ rad/s and $\alpha_i = 18 \text{ s}^{-1}$. Increasing damping reduced overall joint RMSE from 0.5381° to 0.4698° , TCP position RMSE from 6.566 to 5.836 mm, and TCP maximum position error from 25.467 to 20.751 mm. Peak torque utilization remained essentially unchanged.

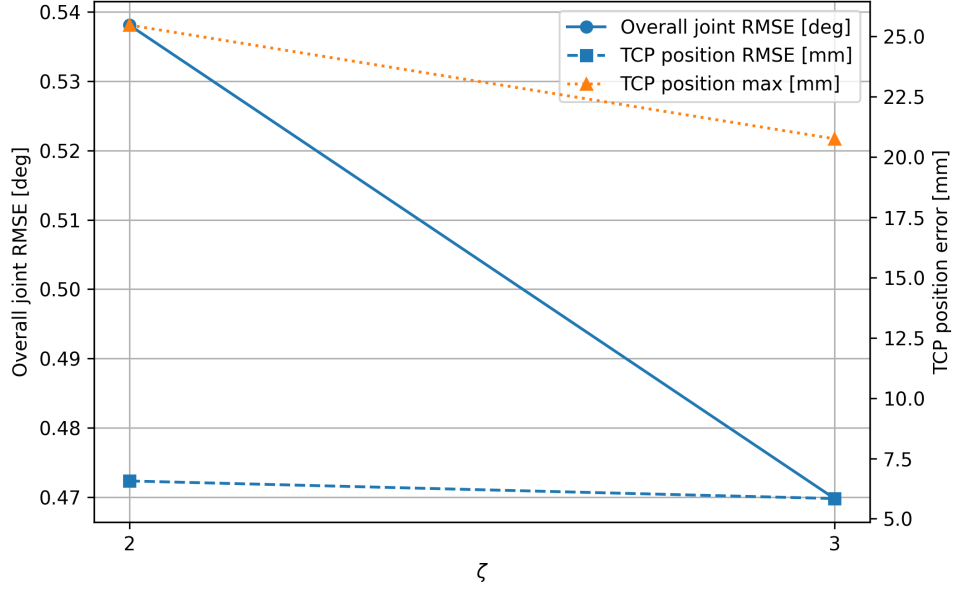


Figure 2: Final damping comparison at $\omega_n = 20$ rad/s and $\alpha_i = 18$ s $^{-1}$.

3.3 Integral action

With $\omega_n = 20$ rad/s and $\zeta = 3$, I retuned $K_i = \alpha_i K_p$. The final local sweep was:

Table 1: Final integral-action sweep.

α_i	Joint RMSE [deg]	TCP RMSE [mm]	TCP max [mm]	TCP final [mm]
14	0.5074	6.330	22.455	0.107
16	0.4861	6.051	21.540	0.194
18	0.4698	5.836	20.751	0.115
20	0.4575	5.668	20.059	0.215

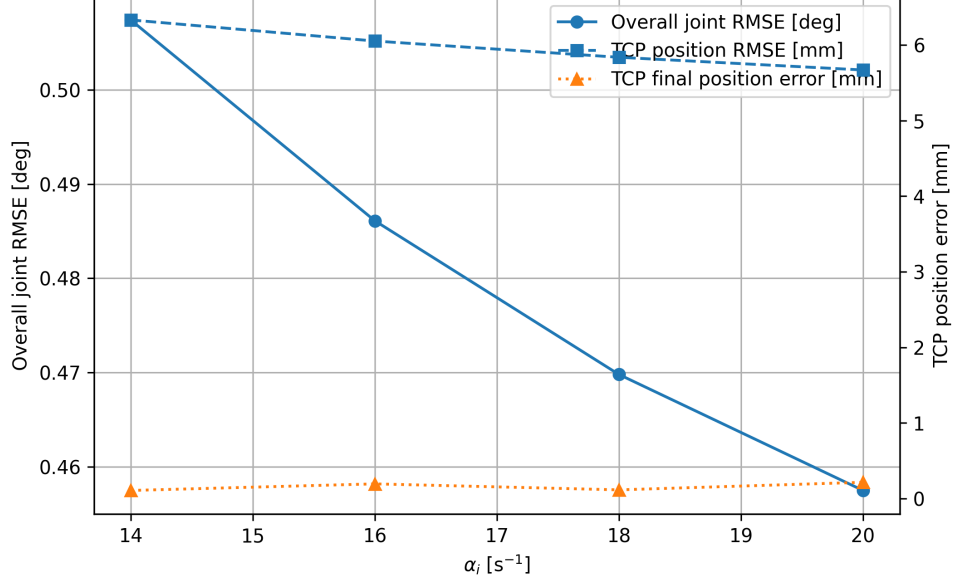


Figure 3: Final integral-gain sweep. $\alpha_i = 20$ slightly reduced trajectory RMSE, but final TCP position error increased relative to $\alpha_i = 18$.

The trajectory-integrated metrics improve consistently as α_i increases over this local range, whereas the final TCP error is non-monotonic. This does not contradict the overall tuning trend. Unlike RMSE, the final error is an instantaneous value evaluated at $t = 4$ s; changing the integral gain alters the transient and settling phase, so the error at that particular instant need not vary monotonically with α_i .

I selected $\alpha_i = 18$. Moving to 20 produced only a modest trajectory-RMSE improvement while final TCP position error increased from 0.115 to 0.215 mm and integral action increased further. The selected value is therefore a practical nominal choice, not a claimed global optimum.

4 Final PID Design

The final parameters are

$$\boxed{\omega_n = 20 \text{ rad/s}, \quad \zeta = 3, \quad \alpha_i = 18 \text{ s}^{-1}}. \quad (14)$$

The frozen diagonal inertia used for gain construction is

$$M_d = [1.05863580, 2.59135828, 0.881358542, 0.0231437242, 0.00503939441, 0.00025756].$$

Table 2: Final joint PID gains.

Joint	K_p	K_d	K_i
J1	423.4543	127.0363	7622.1778
J2	1036.5433	310.9630	18657.7796
J3	352.5434	105.7630	6345.7815
J4	9.2575	2.7772	166.6348
J5	2.0158	0.6047	36.2836
J6	0.1030	0.0309	1.8544

The large numerical variation across joints is intentional and follows the strong variation in effective joint inertia at the selected design configuration.

5 Nominal Benchmark Results

5.1 Joint-space tracking

Table 3: Final PID joint-space tracking.

Joint	RMSE [deg]	Maximum [deg]	Final [deg]
J1	0.00481	0.01047	0.00434
J2	0.11640	0.39698	0.00449
J3	0.31335	1.12948	0.00436
J4	1.09802	3.93675	0.00685
J5	0.08386	0.16098	0.01566
J6	0.00803	0.01631	0.00103

The overall joint RMSE was

$$0.4698^\circ.$$

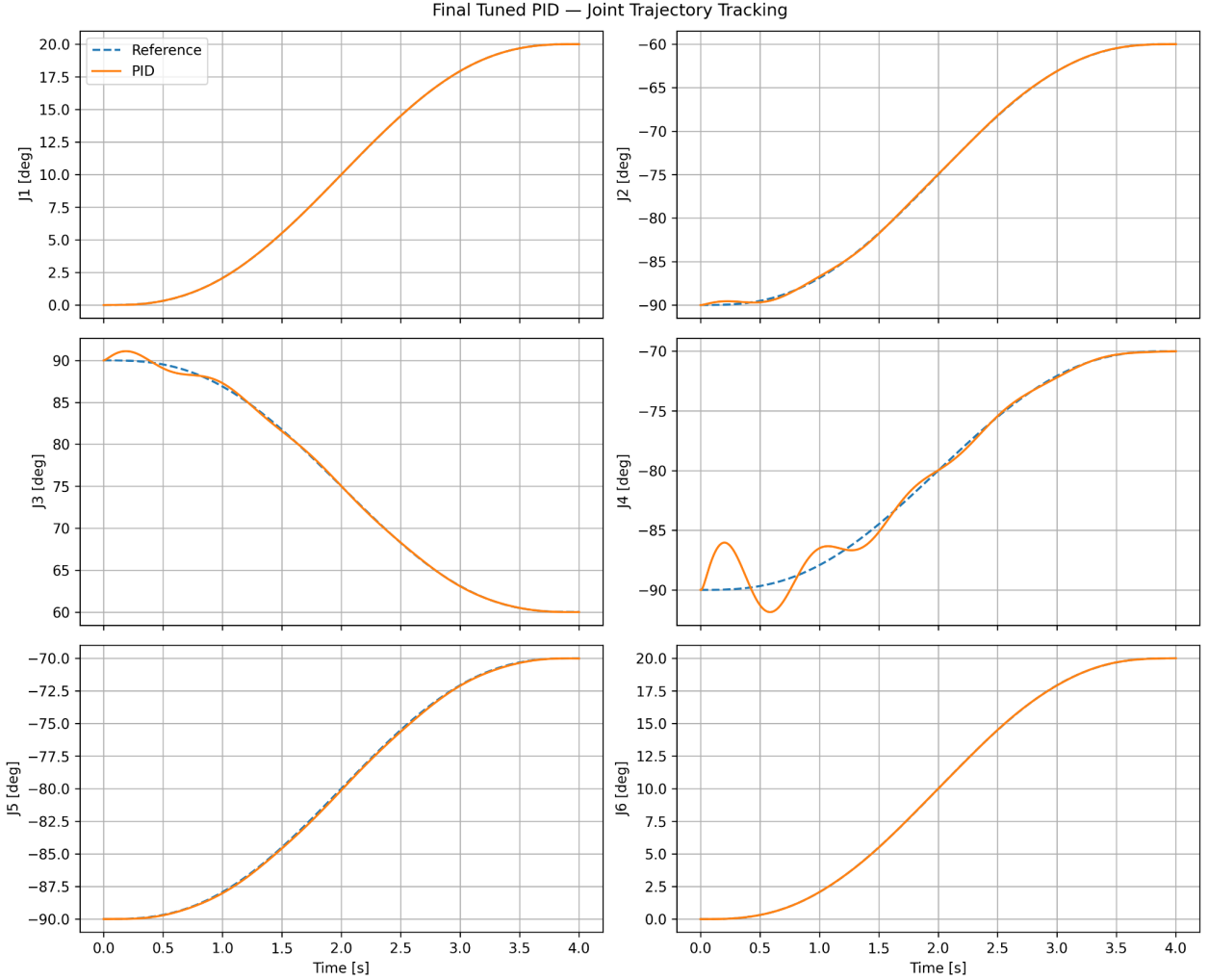


Figure 4: Final PID trajectory tracking. Reference and actual trajectories largely overlap at full-motion scale.

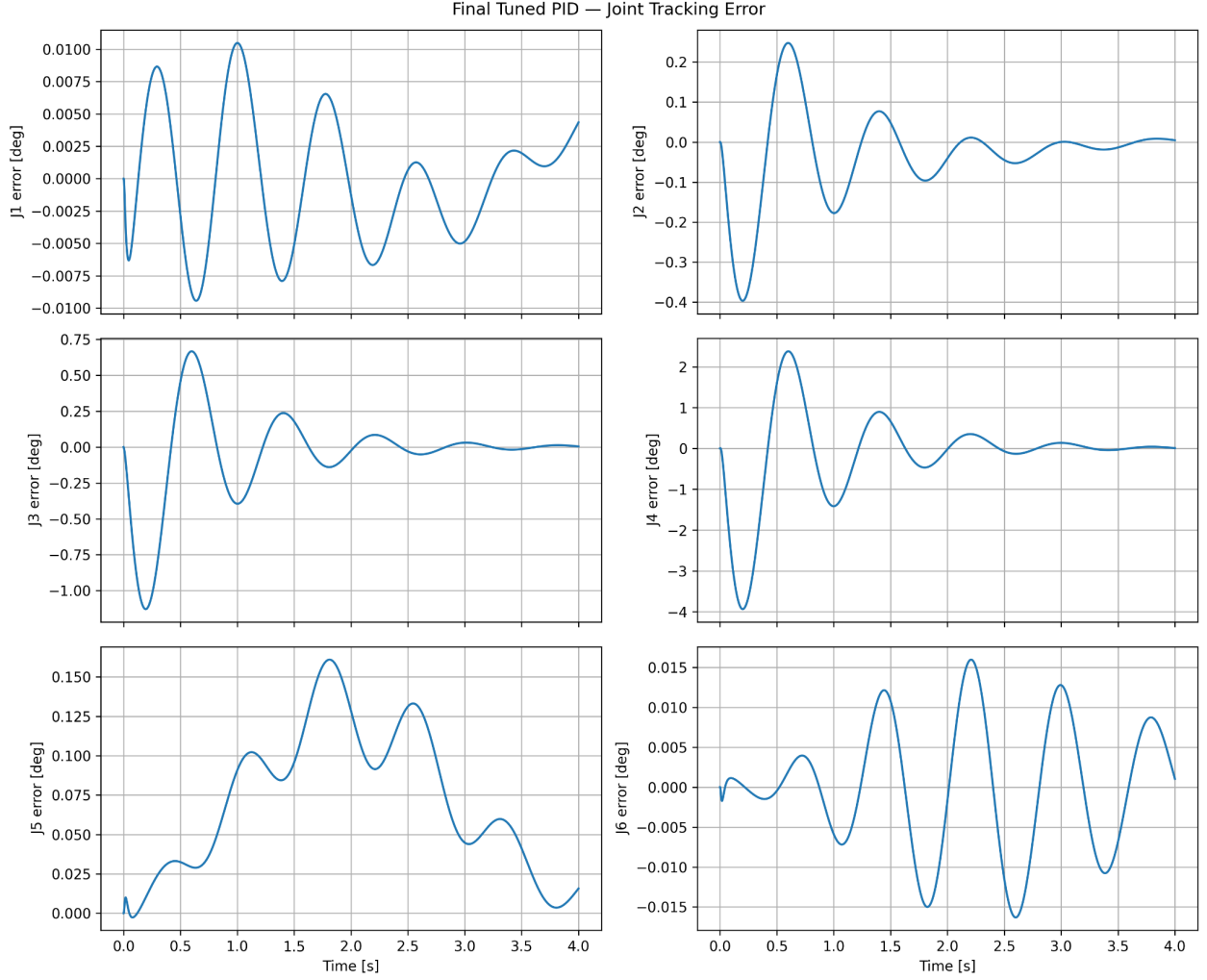


Figure 5: Final PID tracking error. Joint 4 is the dominant trajectory-tracking limitation, although its terminal error is small.

Joint 4 is the clearest limitation. Its RMSE of 1.098° and peak error of 3.937° are much larger than its final error of 0.00685° . The controller therefore converges accurately to the endpoint, while its principal weakness appears during motion. Rather than attributing this behavior generically to nonlinear dynamics, I performed an additional dynamic decomposition to determine which physical terms dominate the transient.

5.2 Dynamic Origin of the Joint 4 Tracking Error

For Joint 4, I decomposed the model-based torque demand as

$$\tau_4^{\text{ID}} = M_{44}(q)\ddot{q}_{4,d} + \sum_{j \neq 4} M_{4j}(q)\ddot{q}_{j,d} + c_4(q, \dot{q}) + g_4(q). \quad (15)$$

The decomposition was evaluated along the actual PID state trajectory $q(t), \dot{q}(t)$ using the desired acceleration $\ddot{q}_d(t)$. This separates the torque associated with Joint 4's own desired acceleration, acceleration-induced inertial coupling from the other five joints, velocity-dependent Coriolis/centrifugal effects, and gravity.

The maximum absolute Joint 4 tracking error occurred at

$$t = 0.200 \text{ s}, \quad e_4 = -3.936753^\circ.$$

At that instant, the individual torque contributions were

$$M_{44}\ddot{q}_{4,d} = +0.001295 \text{ N m}, \quad (16)$$

$$\sum_{j \neq 4} M_{4j}\ddot{q}_{j,d} = -0.002059 \text{ N m}, \quad (17)$$

$$c_4(q, \dot{q}) = +0.000008 \text{ N m}, \quad (18)$$

$$g_4(q) = -1.763143 \text{ N m}. \quad (19)$$

The resulting model-based torque demand was

$$\tau_4^{\text{ID}} = -1.763899 \text{ N m},$$

while the actual PID command was

$$\tau_4^{\text{PID}} = -2.005831 \text{ N m}.$$

The instantaneous difference was therefore approximately

$$\tau_4^{\text{ID}} - \tau_4^{\text{PID}} = 0.241932 \text{ N m}.$$

More importantly, the relative magnitudes remained strongly separated over the entire trajectory. The maximum absolute values were

Table 4: Maximum absolute Joint 4 dynamic torque contributions over the benchmark trajectory.

Dynamic contribution	Maximum magnitude [N m]
Self-inertia	0.002915
Inertial coupling	0.003595
Coriolis/centrifugal	0.006216
Gravity	1.850120

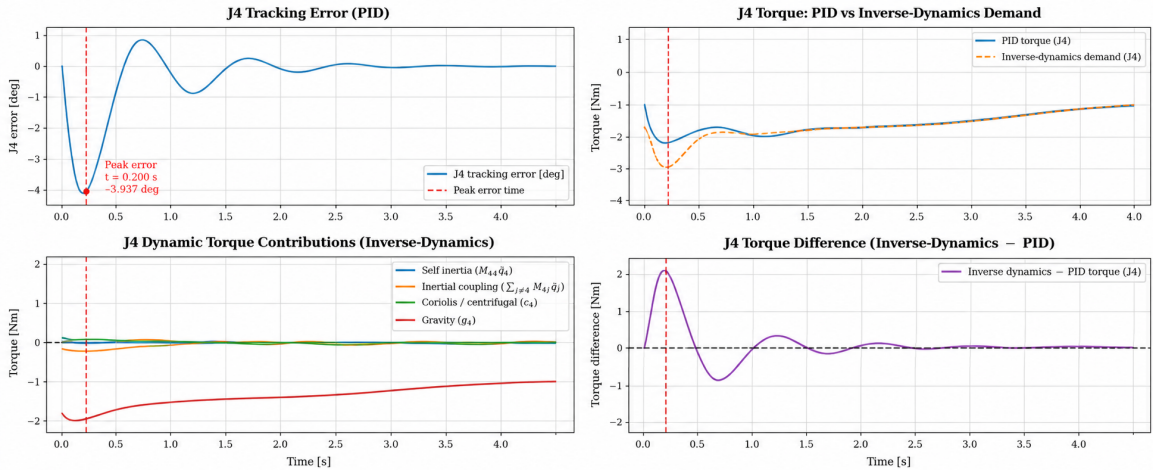


Figure 6: Joint 4 tracking error and inverse-dynamics decomposition along the final PID trajectory. The gravitational contribution dominates the Joint 4 torque demand, whereas self-inertia, inertial-coupling, and Coriolis/centrifugal terms remain comparatively small. The maximum tracking error occurs at $t = 0.200$ s.

The decomposition rejects the initial possibility that Joint 4 is difficult in this benchmark because of unusually strong inertial coupling. The largest inertial-coupling contribution is approximately 0.0036 N m and the largest Coriolis/centrifugal contribution approximately 0.0062 N m, whereas

the gravitational term reaches approximately 1.85 N m. Joint 4 is therefore overwhelmingly gravity-dominated under the present trajectory.

This observation also explains why the transient is especially visible with a feedback-only PID controller. At the initial condition,

$$e(0) = 0, \quad \dot{e}(0) = 0, \quad \int_0^0 e(t) dt = 0,$$

so the PID feedback command initially contains no compensating torque. The plant, however, requires nonzero gravitational torque immediately. At the benchmark initial configuration, the Joint 4 gravitational demand is approximately -1.825 N m. Joint 4 must therefore begin to deviate from the reference before proportional, derivative, and subsequently integral feedback generate the required compensating torque.

The resulting peak error should not be interpreted as insufficient actuator authority. The final benchmark uses only 7.24% of the adopted Joint 4 torque limit. Instead, the transient exposes the reactive nature of the controller: gravitational loading is compensated after it manifests as tracking error rather than being cancelled explicitly through a model-based feedforward term.

5.3 Task-space tracking

Joint error was propagated through forward kinematics. Position error is the Euclidean translational norm; orientation error is the geodesic angle of the relative rotation rather than Euler-angle subtraction.

Table 5: Final PID TCP tracking.

Metric	RMSE	Maximum	Final
Position [mm]	5.836	20.751	0.115
Orientation [deg]	1.528	5.462	0.022

The endpoint is accurate, but the path contains substantially larger Cartesian deviations. For path-critical tasks, terminal error alone would therefore overstate this controller’s tracking quality.

5.4 Control effort

Table 6: Final PID control effort.

Joint	Maximum [N m]	RMS [N m]	Utilization [%]
J1	0.6435	0.1308	0.43
J2	40.7535	31.6801	27.17
J3	22.1596	20.1451	14.77
J4	2.0282	1.6916	7.24
J5	0.1802	0.1131	0.64
J6	0.00011	0.00002	≈ 0

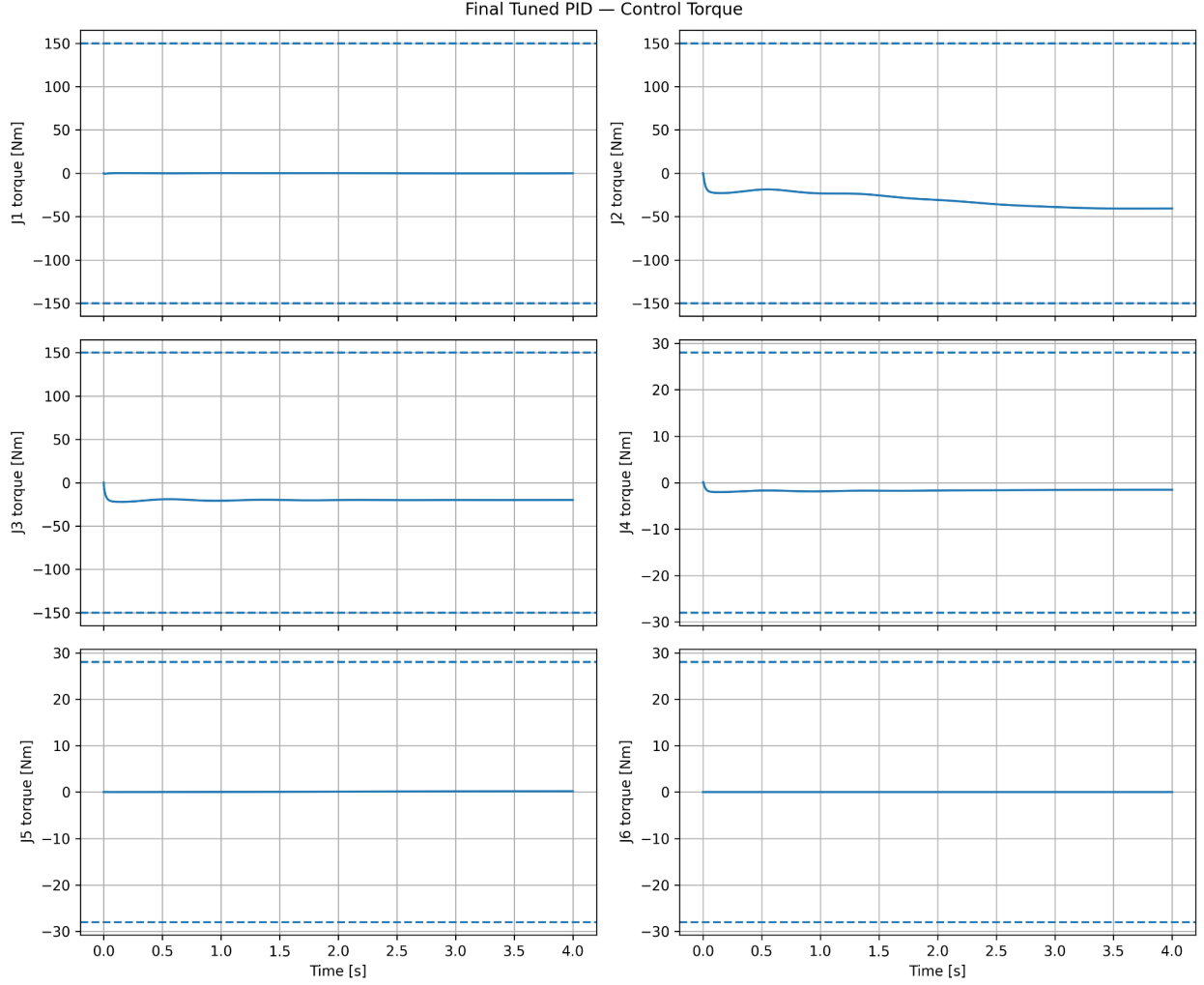


Figure 7: Final PID torque histories and adopted simulation constraints. No joint approaches saturation.

Joint 2 carries the dominant absolute torque demand, but its peak uses only 27.17% of the adopted limit. Joint 4 reaches only 7.24% utilization. The nominal benchmark is therefore not saturation dominated, and the residual trajectory error cannot be explained by insufficient modeled torque authority.

5.5 Python–C++ Implementation Cross-Validation

Before closing the PID baseline, I independently implemented the final controller in C++¹⁷ and cross-validated it against the Python reference implementation. The purpose was not to repeat the nonlinear benchmark in a second language, but to verify that the controller equations, gain construction, state update, saturation logic, and anti-windup behavior were reproduced consistently in an implementation suitable for later ROS 2 integration.

Both implementations used the final design parameters $\omega_n = 20$ rad/s, $\zeta = 3$, and $\alpha_i = 18$ s⁻¹, with the same 2 ms control period and joint torque limits. Four deterministic test cases were evaluated:

1. zero tracking error;
2. a 1° Joint 2 position error;
3. simultaneous position and velocity errors on all six joints;

4. repeated large errors to exercise torque saturation and conditional-integration anti-windup.

The final Python and C++ implementations produced matching controller outputs in all four cases. For example, the 1° Joint 2 test produced

$$\tau_2 = 18.742373005 \text{ N m}$$

in both implementations. The mixed-error test also reproduced the same six-axis torque vector,

$$\tau = [10.738801555, 9.619121935, -5.097777807, -0.148860434, 0.082444493, 0.000787103] \text{ N m}.$$

Under the repeated-saturation test, both implementations converged to

$$\tau = [150, 150, 150, 28, 28, 28] \text{ N m},$$

with the same integral state,

$$e_I = [0, 0, 0, 0, 0.22, 14.56].$$

The cross-validation also exposed an implementation-level distinction during development: an initial C++ version evaluated the anti-windup condition using a candidate updated integral state, whereas the Python reference evaluated saturation using the current integral state before performing the conditional update. This produced a one-sample difference in the Joint 5 and Joint 6 integral states. After the C++ update sequence was corrected to reproduce the reference algorithm, all deterministic test outputs matched.

This cross-validation provides an implementation-consistency check between the Python analysis controller and its C++ counterpart. It does not constitute an independent validation of the PID control law or the physical UR5e plant; those questions are addressed separately through the nonlinear benchmark and later simulation validation.

6 Computation Cost

The measured controller-only execution times in the current Python implementation were

$$t_{\text{mean}} = 0.024661 \text{ ms}, \quad t_{\text{max}} = 0.051881 \text{ ms}.$$

These are development-computer measurements of the implemented controller function, not UR5e hardware latency, ROS 2 end-to-end latency, or proprietary UR servo timing. Their purpose is consistent relative comparison with later controllers. Relative to the 2 ms simulation control period, PID computation is negligible.

7 Engineering Interpretation

The study supports four main observations.

First, fixed-gain PID can regulate the final configuration accurately in the nominal model without explicit gravity or nonlinear feedforward compensation. Integral action removes most residual endpoint error, but it does not make the controller dynamically equivalent to a model-based compensator.

Second, endpoint regulation and trajectory tracking are different problems. Joint 4 ends within 0.00685° yet reaches 3.937° error during motion. TCP position likewise ends at 0.115 mm error while reaching 20.751 mm during the trajectory. A controller judged only from final error would therefore appear substantially better than it is as a trajectory-tracking controller.

Third, the dominant Joint 4 transient is not caused by actuator saturation or unusually large inertial coupling. The additional inverse-dynamics decomposition showed that Joint 4 is overwhelmingly gravity-dominated in this benchmark: the gravitational contribution reaches approximately

1.85 N m, whereas inertial-coupling and Coriolis/centrifugal terms remain below approximately 0.004 and 0.007 N m, respectively.

This distinction is important. The fixed-gain PID controller has no explicit gravity feedforward term, so the required gravitational compensation must emerge through tracking error and the resulting feedback action. Joint 4 therefore provides a particularly clear example of the difference between reactive feedback compensation and model-based dynamic compensation.

Fourth, indefinitely increasing gains would not improve the quality of the controller comparison. Once the selected design range had been explored and the remaining behavior was physically characterized, I fixed the PID baseline rather than turning the study into a search for a numerical failure boundary. This preserves PID as an interpretable baseline against which more model-dependent controllers can be evaluated.

8 Limitations

- The plant is an ideal fixed-base rigid-body UR5e model with no external payload.
- Friction, backlash, compliance, actuator electrical/gearbox dynamics, sensor noise, communication delay, and the proprietary UR servo are not modeled.
- The adopted torque values are simulation constraints carried from the imported model, not independently verified motor ratings.
- The controller is tuned and evaluated against the same nominal model. Payload, parameter mismatch, and external disturbance are reserved for the common robustness comparison.
- ω_n and ζ are design parameters based on a frozen diagonal inertia approximation, not exact global nonlinear closed-loop modal properties.
- The Joint 4 inverse-dynamics decomposition evaluates model-based torque demand along the actual PID state trajectory using the desired acceleration. It is intended to identify the relative magnitude of modeled dynamic contributions, not to reconstruct the actual closed-loop acceleration or prove causality from torque difference alone.
- Controller timing is implementation- and hardware-dependent and is used primarily as a relative metric.

9 Conclusion and Transition

The final PID baseline provides accurate endpoint regulation and reasonable nominal trajectory tracking without approaching the modeled actuator limits. Its principal weakness is motion-dependent error, most clearly exposed by Joint 4.

Additional dynamic decomposition showed that this behavior is not primarily a coupling or saturation problem. Under the present trajectory, Joint 4 is strongly gravity-dominated. Because the feedback-only PID controller has no explicit gravity compensation, it must first develop tracking error before generating the torque required to oppose the gravitational load. The small final error therefore coexists with a substantially larger transient error.

This finding provides a direct physical motivation for the next controller. Computed torque introduces explicit $M(q)$, $c(q, \dot{q})$, and $g(q)$ compensation together with reference acceleration. The next stage can therefore test whether the Joint 4 transient identified here disappears when the dominant model-based torque demand is supplied proactively rather than generated reactively through tracking error.

The broader comparison will not ask only which controller produces the smallest nominal error. It will also examine what is exchanged for improved tracking in terms of model dependence, computational burden, implementation complexity, and robustness to mismatch.